Integrated Pest Management

Agriculture RAG Knowledge Base • Educational reference document

Integrated Pest Management (IPM) is an approach that combines monitoring, prevention, biological methods, cultural practices, mechanical methods, and chemical control when appropriate. The goal is to manage pests while reducing unnecessary pesticide use and protecting beneficial organisms.

A basic IPM process begins with identification. Farmers should determine whether the observed organism is actually a pest, distinguish pests from beneficial insects, and understand the crop stage affected. Monitoring can involve field scouting, traps, visual counts, or other appropriate techniques.

Preventive practices include crop rotation, sanitation, resistant or tolerant varieties where available, balanced nutrition, appropriate planting dates, and habitat management. Biological control uses natural enemies such as predators or parasitoids. Mechanical methods may include physical removal or barriers.

Chemical control can be part of IPM when other measures are insufficient and when an appropriate product is legally registered and used according to its label. Application should consider target pest, crop, timing, dose, weather, resistance management, worker safety, and environmental risks.

A key principle is that not every insect or symptom requires pesticide treatment. Decisions should be based on monitoring and locally appropriate thresholds or recommendations. Integrated approaches can reduce unnecessary inputs while maintaining crop protection.

Reference basis

• FAO, Plant Production and Protection resources

• FAO, pesticide management materials

Note: This document is a project knowledge-base document prepared for a student RAG demonstration. It is not a substitute for crop-specific agronomic advice, soil testing, product labels, or local agricultural recommendations.